

Transforming Shopping Experience

Through Market Basket Insights

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Executive Summary

- In late 2015, Walmart experienced a 17% sales decline, signaling a potential drop in customer satisfaction. This prompted the company to reassess its customer shopping experience using a data-driven approach.
- The objective of the project was to identify product combinations that would allow for personalized recommendations, bundled promotions, and optimized product placements, all aimed at enhancing the customer experience and driving sales growth.
- Market Basket Analysis was applied to uncover relationships between products that were frequently bought together, providing actionable insights for product placement and promotions.
- Key findings included several cross-selling recommendations. For example, more than 40% of customers who purchased Potato Products also bought Beef, as well as similar patterns with Canned Fruit & Coffee, and Meat Spreads & Eggs. Bundling these products can lead to a positive impact on sales.
- Other combinations, such as Kitchen Towels & Hut-Milk, Rice & Napkins, Sparkling Wines & Waffles, and Flour & Mayonnaise, showed that more than 23% of customers bought both items. To leverage this insight, placing these products closer together on shelves would improve cross-selling and customer experience.



Company Background

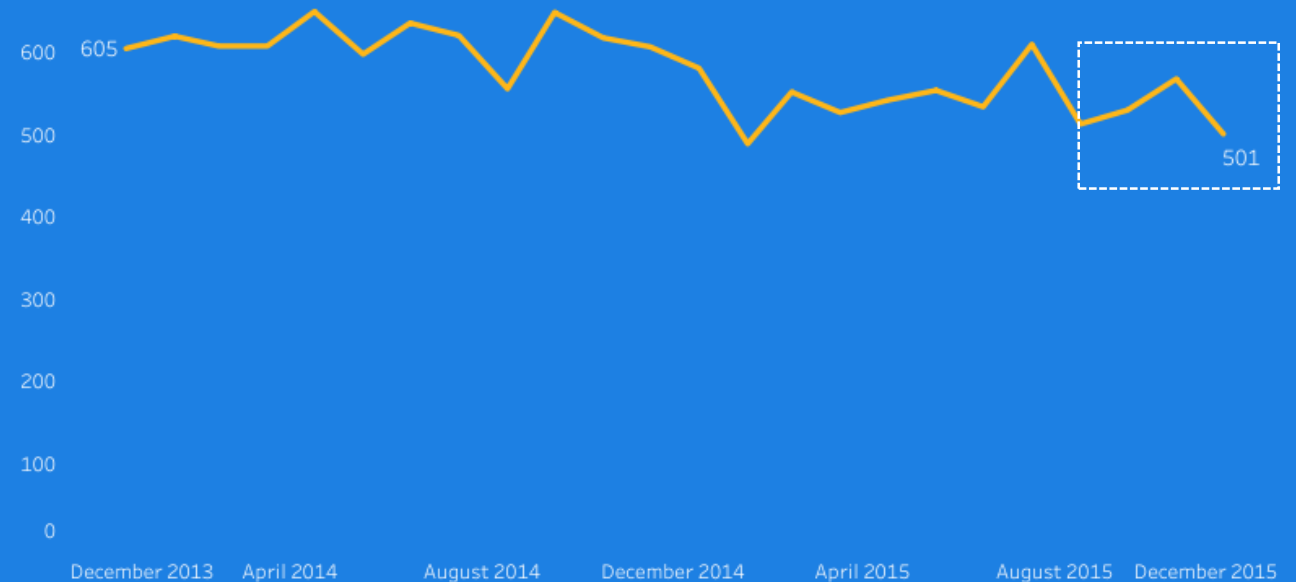


Walmart, a global retail giant, has been a leader in the retail industry for decades, known for its commitment to providing a wide variety of products at affordable prices. With millions of customers visiting both physical stores and online platforms, Walmart continually seeks innovative ways to enhance the shopping experience and maintain its competitive edge.



Problem Statement

In late 2015, Walmart noticed a slight decline in sales by 17%, signaling a potential drop in customer satisfaction. This trend highlighted the need to better understand customer behavior and preferences. The challenge is to leverage data-driven insights to create a more personalized and engaging shopping experience, ensuring that customers find value in each visit and are encouraged to return more frequently.





Problem Objective

Identify the best product combinations to propose personalized recommendations, bundled promotions, and optimized product placements that enhance the shopping experience and drive sales growth.

Root Cause Analysis



Dataset Overview

Transactions

38,008

Type of Products

167

Period

2014 - 2015

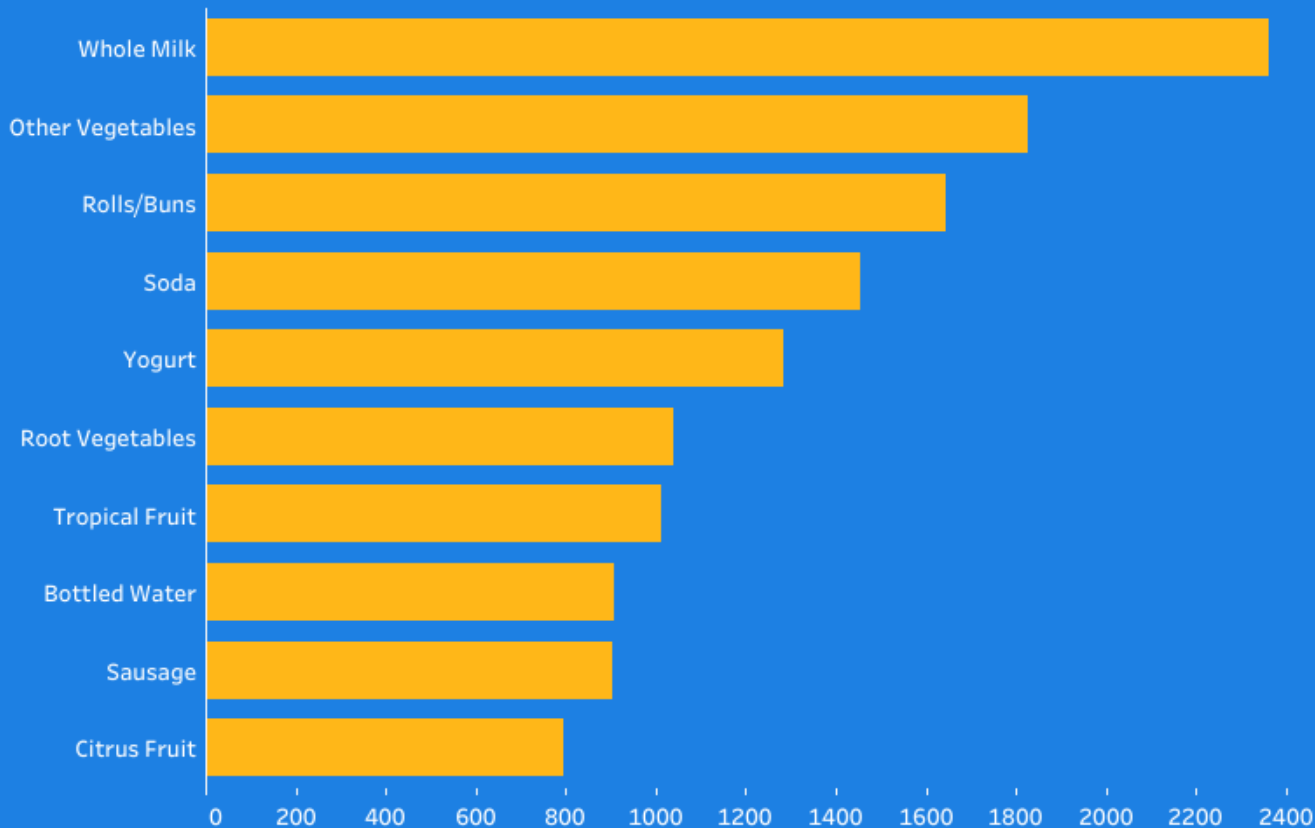


A photograph of a Walmart store interior, overlaid with a semi-transparent blue filter. The image shows various retail displays, including a large M&M's character cutout in the foreground, a tall display of Reese's Peanut Butter Cups on the right, and a customer service counter in the background. The Walmart logo and slogan are visible on a wall in the background.

Exploratory Data Analysis

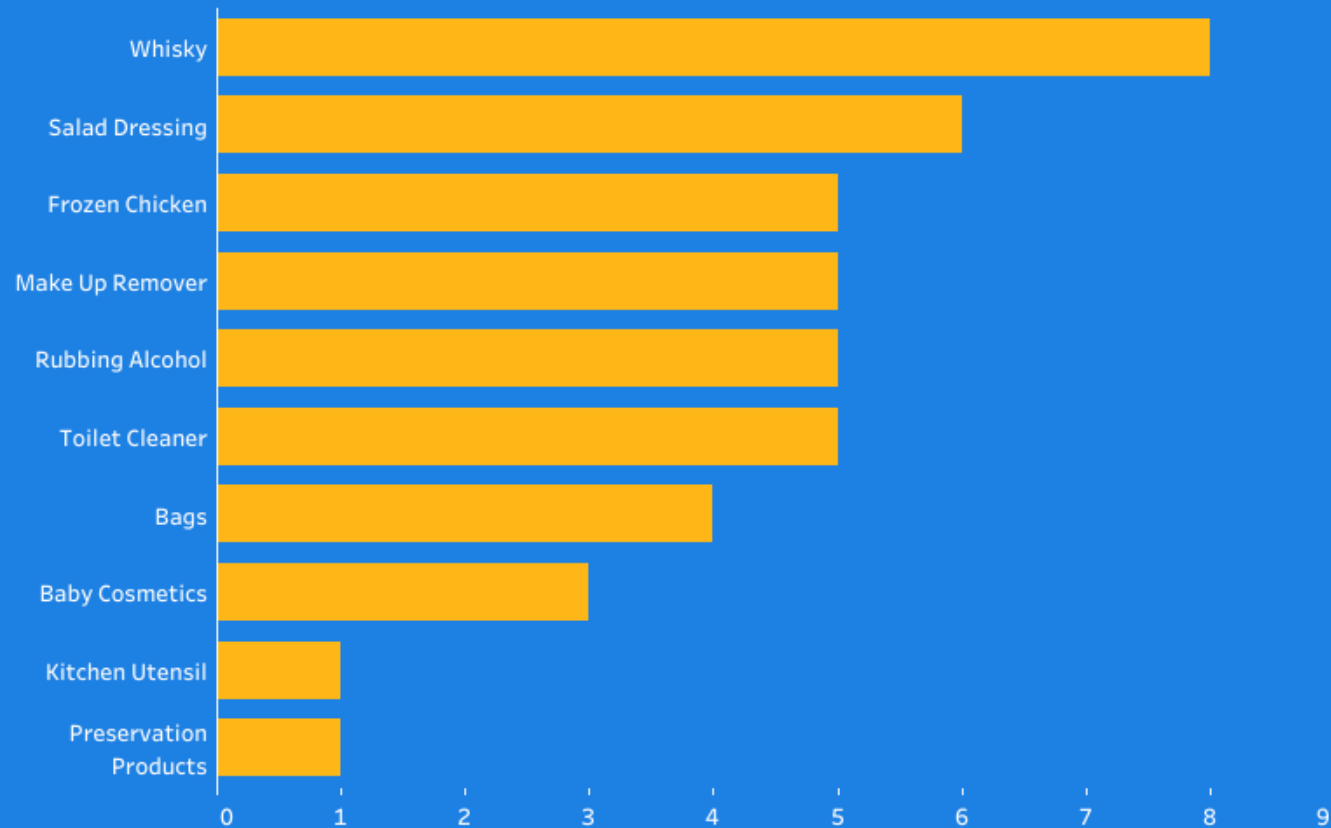
Top 10 Most Popular Products (2014 – 2015)

Customers primarily purchase milk, vegetables, fruits, and ready-to-eat (RTE) foods. The company can enhance both customer experience and sales by offering these popular items in bundles.



Top 10 Least Popular Products (2014 – 2015)

These products sold less than 8 items in two consecutive years, which make the company lose. Since the company buy these products but failed to gain profit.



A blue-tinted photograph of a grocery store aisle. In the foreground, several large black plastic bins are filled with bright yellow lemons. The bins are arranged in a row, receding into the background. In the background, other shoppers are visible, including a woman pushing a shopping cart filled with various items. A sign above a section of the store reads "Deli". Another sign in the background says "Everyday Low Price". The overall scene is a typical grocery store environment.

Market Basket Analysis

3 Metrics in Market Basket Analysis

S

C

L

Support

Confidence

Lift

How Popular an item set
is among the customer.

How likely an item Y is
purchased when item Xi
is purchased.

The ratio between
target response and
average response.



By analyzing key metrics such as **Support** (minimum of 0.2%), **Confidence** (minimum of 5%), and **Lift** (minimum of 3), we can identify items that are frequently bought together. By leveraging these insights, we will combine these associated items to enhance product sales and create a more engaging shopping experience. This targeted approach will enable Walmart to personalize recommendations, optimize product placements, and design promotions that resonate with individual customers, ultimately driving customer satisfaction and reversing the sales decline.



Apriori Algorithm Result

Based on the 3 metrics criteria, there are 7 product combinations with the highest association and high influence to boost the products' probability to be bought.

	1st Product	2nd Product	Support	Confidence	Lift
1	Potato Products	Beef	0.002565	0.454545	3.802185
2	Canned Fruit	Coffee	0.002309	0.428571	3.728954
3	Meat Spreads	Domestic Eggs	0.003592	0.4	3.004239
4	Kitchen Towels	Hut-Milk	0.002309	0.3	3.821569
5	Rice	Napkins	0.003079	0.244898	3.011395
6	Sparkling Wine	Waffles	0.002565	0.217391	3.150154
7	Flour	Mayonnaise	0.002309	0.06338	3.338599



Cross Selling Recommendation

1. Potato Products & Beef

2. Canned Fruit & Coffee

3. Meat Spreads & Eggs

More than **40%** of the customer who buy the 1st product will also purchase the 2nd item. These combination will bring positive impact to the sales as shown in high lift score.



Cross Selling Recommendation

4. Kitchen Towels & Hut-Milk

5. Rice & Napkins

6. Sparkling Wines & Waffles

7. Flour & Mayonnaise

More than **23%** of the customer who buy the 1st product will also purchase the 2nd item. Besides selling them as bundling product, the company should put these item close on the shelf.



A photograph of a Walmart store interior, overlaid with a semi-transparent blue filter. The image shows various retail displays, including M&M's character cutouts and shelves of products. In the background, a sign for 'Service à la clientèle' (Customer Service) is visible. The Walmart logo and slogan 'Économisez plus.' are prominently displayed on a wall in the center-right.

Dashboard

Sales
\$112K



▲+13.4% vs. PY

Profit
\$16K



▲+90.6% vs. PY

Orders
1,092



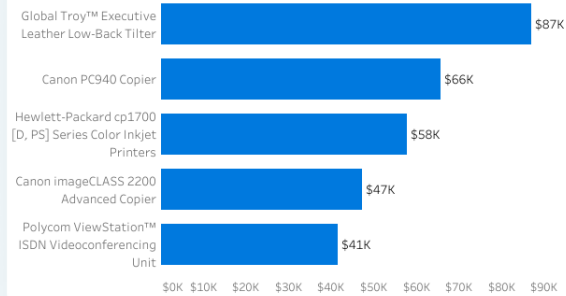
▼-28.0% vs. PY

Profit Margin
14.0%

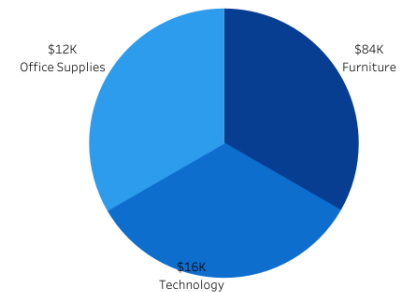


▲+68.2% vs. PY

Performnce By Product



Performance By Product Category



Performance By Product Sub-Category

Product Sub-Category	Sales	Profit	Orders
Appliances	\$4.66K	\$0.95K	109
Binders and Binder Acces...	\$2.66K ▼	\$0.43K ▼	132 ▼
Bookcases	\$4.58K ▲	(\$1.62K) ▲	47 ▲
Chairs & Chairmats	\$49.45K ▲	\$14.48K ▲	117 ▲
Computer Peripherals	\$6.75K ▼	\$1.69K ▼	118 ▼
Copiers and Fax	\$1.29K ▼	(\$1.57K) ▼	3 ▼
Envelopes	\$0.77K ▼	\$0.40K ▼	46 ▼
Labels	▼	▼	▼
Office Furnishings	\$2.45K ▲	\$0.37K ▲	176 ▲
Office Machines	\$3.90K ▲	\$1.15K ▲	40 ▲
Paper	\$2.07K ▼	\$0.57K ▼	47 ▼
Pens & Art Supplies	\$1.09K ▼	\$0.15K ▼	26 ▼
Rubber Bands	▼	▼	▼
Scissors, Rulers and Trim...	▼	▼	▼
Storage & Organization	\$0.30K ▲	(\$0.04K) ▲	24 ▲
Tables	\$27.64K ▲	(\$1.78K) ▲	109 ▲
Telephones and Communi...	\$4.37K ▼	\$0.44K ▼	98 ▼
Grand Total	\$111.98K	\$15.63K	1,092

Thank You

If you have any questions or would like to discuss the insights further, feel free to connect with me on [LinkedIn](#).